Pure subfield geometries and a scalar obstruction

Human Compression project

For standard simple subfield parabolics, the exact criterion for residual weak primitivity is that their common subfield remain simple. Descent proves that the resulting geometries are thick, residually connected and flag-transitive. In the mixed parabolic of Construction 10, maximality is instead equivalent to a scalar codimension of one; this disproves the source claim for every prescribed parameter.

1 Exact source scope and retained inputs

Write the odd field exponent as

$$n = \prod_{j=1}^{m} p_j^{f_j}, \qquad p_i \neq p_j \ (i \neq j).$$

Thus m counts distinct prime factors, whereas $\Omega(n) = \sum_j f_j$ counts factors with multiplicity. Construction 10, printed p.18, specifies rank $3 \leq \alpha \leq \min(m+1, \Omega(n))$, selects $k = \alpha - 1$ pairwise nonisomorphic maximal Suzuki subfield groups, puts $a_i = n/\log_2(s_i)$, and sets $t = 2^{n/\prod_i a_i}$. Its extra parabolic has type $(E_q \hat{\cdot} E_t) : C_{t-1}$. Construction 11 instead specifies $2 \leq \alpha \leq m-1$ and selects α pairwise nonisomorphic maximal Suzuki subfield groups. Theorem 5 immediately below claims that Constructions 1–11 give all RWPRI pre-geometries of rank at least four. The rank bound and the distinct-prime convention were checked in the page image, not only its text extraction.

The source's definition on pp.4–5 is the exact condition used here: for every proper subset J of the type set, at least one remaining intersection $G_{J\cup\{i\}}$ must be maximal in G_J. In particular, the condition applies to rank-one residues. The hat-dot on p.5 means a nonsplit extension; E_q has order q. The introduction to the constructions on p.15 instructs the reader to retain their whole intended intersection lattice, inferred from the rank-four discussion. Construction 10 comes from the R12 discussion on pp.14–15. Its intersections with the selected subfield groups have type $(E_{s_i} \hat{\cdot} E_t) : C_{t-1}$.

We retain the following classical group theory.

1. The Suzuki realization inside Sp_4, with exceptional Steinberg endomorphism ρ, $\rho^2 = \phi_2$, and compatible standard groups $S_d = \mathrm{Sz}(2^d) = \mathbf{H}^{\rho^d}$ for positive odd d. Here $S_1 = \mathrm{Sz}(2) \cong 5:4$ is allowed as an intersection, although it is not simple.
2. Prime-relative-degree inclusions $S_d < S_{ad}$, for odd prime a and $d > 1$, are maximal. Standard subfield intersections are given by the gcd of their exponents.
3. The Suzuki Sylow-2 group U_q has exponent four and its identity together with its involutions is $Z_q = Z(U_q)$, elementary abelian of order q. A nonidentity involution $z \in Z_q$ has $C_G(z) = U_q$. Consequently a normalizer of Z_q normalizes U_q, while a normalizer of U_q normalizes its center: $N_G(Z_q) = N_G(U_q) = B_q$, the ovoid point stabilizer $U_q : H_q$, where $H_q \cong C_{q-1}$.
4. Nonsplit torus normalizers have order $4(q \pm \sqrt{2q} + 1)$ and are maximal for $q \geq 8$. Odd-element centralizers and these normalizers have their usual Suzuki structure.

5. Schur–Zassenhaus complement conjugacy for the solvable group $U_q : H_t$. This puts its subgroups of order $t-1$ into the standard H_t.
6. Lang–Steinberg surjectivity on the connected algebraic group Sp_4, as used in the proved finite-descent and coset-Helly statements of Field descent and compatible tori, Section 2.

The matrix action used in Section 4 is verified directly below. The retained facts do not include any source classification of all RWPRI lattices.

2 Sharp criterion for the canonical pure-subfield family

Theorem A. Let n be positive and odd. Select $k \geq 2$ distinct prime divisors $a_1, \ldots, a_k$ of n, put

$$e = \frac{n}{\prod_{i=1}^{k} a_i}, \qquad G = S_n, \qquad G_i = S_{n/a_i}.$$

The standard coset incidence system $\Gamma(G; G_1, \ldots, G_k)$ is a thick, residually connected, flag-transitive geometry. It is RWPRI **if and only if** $e > 1$. When $e > 1$, every one-step inclusion in its intersection lattice, not only one inclusion at each subset, is maximal.

Proof. For $J \subseteq \{1, \ldots, k\}$, set

$$n_J = \frac{n}{\prod_{j \in J} a_j}.$$

Entrywise field intersection gives $G_J = S_{n_J}$, including $G_\varnothing = G$. Apply descent Theorem C with $F = \rho^e$ and $m = \prod_i a_i$. Any pairwise incident family of cosets has a common representative. It therefore extends to a translate of the standard chamber. This proves the geometry axiom and flag transitivity.

Suppose that a residue has at least two remaining types. For distinct remaining i, j, the two residue parabolics S_{n_J/a_i} and S_{n_J/a_j} are distinct maximal subgroups of S_{n_J}: their exponents exceed one, even if $e = 1$, because at least one other relative prime remains. They generate S_{n_J}. The standard coset-graph connectivity criterion therefore proves that residue connected. Flag transitivity identifies every residue with such a standard one, proving RC.

The panel of type i has $[S_{ea_i} : S_e]$ chambers. This is at least three. Indeed, S_{ea_i} is nonabelian simple and S_e is a proper nontrivial subgroup, so its index cannot be two. This includes $e = 1$, where $|S_e| = 20$. Thus the geometry is thick.

If $e > 1$, each remaining exponent n_J/a_i is at least $e > 1$. Every one-step inclusion is a prime-degree inclusion of simple subfield groups, and hence maximal. This proves RWPRI.

If $e = 1$, the distinct primes a_i, with $k \geq 2$, include at least one $a_i \geq 5$. At the subset containing every type except i, the only remaining inclusion is

$$S_1 < S_{a_i}.$$

Let C_5 be the normal subgroup of $S_1 = 5 : 4$. Inside S_{a_i}, its centralizer is a nonsplit torus. The order-four complement of S_1 normalizes this centralizer, so S_1 lies inside the full torus normalizer. Its torus order is one of $2^{a_i} \pm 2^{(a_i+1)/2} + 1$, both of which exceed five for $a_i \geq 5$. Hence this containment is strict, while the torus normalizer is proper in S_{a_i}. Consequently S_1 is not maximal, and RWPRI fails at that rank-one residue. □

The theorem concerns the standard simultaneous embeddings. It does not assert that arbitrary conjugates of the same maximal-subgroup types have the same intersections or are flag transitive. If another family has a common prescribed simple subgroup S_e with $e > 1$, however, the retained split-dihedral normalizer/transporter argument in Earlier subfield and

scalar proofs, Section B, forces simultaneous standard position after conjugating S_e. Thus within that prescribed common-subfield lattice there is only one ambient conjugacy class for a fixed labelled selection of exponents.

3 Construction 11 and its omitted boundary

Construction 11 has $k \le m-1$, so an unselected prime remains in e. In particular $e > 1$, and Theorem A supplies canonical realizations for **every literal admissible parameter**. It also answers the source's separate flag-transitivity question for this family. It does not establish flag transitivity of all arbitrary relative positions sharing only the maximal-parabolic isomorphism types.

The exact corrected parameter condition for the canonical pure-subfield family is

$$2 \le k \le m, \qquad \frac{n}{\prod_{i=1}^{k} a_i} > 1. \tag{1}$$

For squarefree n, this recovers $k \le m-1$. If any prime factor of n is repeated, it also permits $k = m$. The source's bound omits this latter boundary.

For example take

$$n = 3^2 \cdot 5 \cdot 7 \cdot 11 = 3465, \qquad (a_1, a_2, a_3, a_4) = (3, 5, 7, 11).$$

Then

$$(G_1, G_2, G_3, G_4) = (S_{1155}, S_{693}, S_{495}, S_{315}), \qquad G_{1234} = S_3 = \mathrm{Sz}(8).$$

Theorem A proves this is a rank-four thick, RC, FT, RWPRI geometry, without enumerating its enormous ambient group. Here $m = 4$, whereas Construction 11 allows only ranks two and three. Constructions 1–10 each have at least one soluble maximal parabolic, while all four parabolics here are nonabelian simple. Consequently this stated-action subgroup family is absent from all eleven literal constructions.

This contradicts the **literal exhaustive list** in Theorem 5, as a theorem about the subgroup systems specified in the paper. It is a boundary omission, distinct from proving that no natural extension of the constructions could produce the example: allowing appropriate truncations of larger constructions or replacing Construction 11's bound by (1) would include it. The statement does not classify all unlabelled geometries or all possible actions of G.

4 A scalar-center normal form

Here n and e are odd, $e \mid n$, $e > 1$, $q = 2^n$, and $t = 2^e$. Use standard Suzuki unipotents $U(a, b)$, with $Z_q = \{U(0, b) : b \in \mathbb{F}_q\}$, and diagonal matrices

$$h(\lambda) = \mathrm{diag}(\lambda^{\theta+1}, \lambda, \lambda^{-1}, \lambda^{-\theta-1}), \qquad \theta = 2^{(n+1)/2}.$$

Multiplying these diagonal and unipotent matrices gives

$$h(\lambda)^{-1} U(a, b) h(\lambda) = U(\lambda^{\theta} a, \lambda^{\theta+2} b). \tag{2}$$

Both exponents are coprime to $q-1$. For $\theta+2$, if an odd divisor c divided both $q-1$ and $2^{(n-1)/2}+1$, squaring the latter congruence would give $2^{n-1} \equiv 1 \pmod{c}$, whereas $2^n \equiv 1 \pmod{c}$; thus $2 \equiv 1 \pmod{c}$, forcing $c = 1$. Since $\theta + 2 = 2(2^{(n-1)/2} + 1)$, the claim follows. In particular the standard H_t acts on both the first unipotent parameter and the center by all nonzero scalars in $\mathbb{F}_t$. Any additive invariant subgroup is therefore an $\mathbb{F}_t$-subspace.

Lemma B: normal form. Every subgroup $Q \leq S_n$ of type $(E_q \widehat{\cdot} E_t) : C_{t-1}$ is ambient-conjugate to

$$Q_0 = Z_q U_t H_t. \tag{3}$$

Proof. Its normal Sylow-2 subgroup P has order qt and contains the elementary abelian kernel of order q. Place P inside an ambient U_q. All its involutions lie in Z_q, so that elementary abelian kernel must equal Z_q. Moreover Z_q is exactly the identity together with the involutions of P; it is characteristic in P and normal in Q. Therefore $Q \leq N_G(Z_q) = B_q$.

Project to $B_q/U_q = H_q$. The odd complement of Q maps isomorphically to the unique subgroup of order $t-1$, namely H_t. Complement conjugacy in $U_q : H_t$ allows an ambient U_q-conjugation putting it equal to H_t. Now P/Z_q is an additive subgroup W of the first unipotent parameter, of order t, normalized by H_t. By (2), $W = v\mathbb{F}_t$ for some nonzero v. A diagonal conjugation, which centralizes H_t, scales this line to $\mathbb{F}_t$. Because P contains the full center, it is the full preimage of that line in U_q; thus $P = Z_q U_t$. This proves (3). □

Lemma C: exact maximality test. Suppose $H < Q_0$ has odd part $t-1$ and a normal Sylow-2 subgroup which is not elementary abelian. Then H surjects onto Q_0/Z_q. Put $W = H \cap Z_q$, regarded as an $\mathbb{F}_t$-subspace of $\mathbb{F}_q$. The overgroups of H in Q_0 correspond exactly to the $\mathbb{F}_t$-subspaces containing W, by

$$V \longmapsto Z_V H, \qquad L \longmapsto L \cap Z_q. \tag{4}$$

In particular, H is maximal in Q_0 exactly when $\dim_{\mathbb{F}_t}(\mathbb{F}_q/W) = 1$.

Proof. A non-elementary Sylow-2 subgroup inside U_q contains an element of order four. Its image modulo Z_q is nonzero, so the image of H in Q_0/Z_q has a nonzero normal additive subgroup. The odd projection has full order $t-1$ and acts irreducibly on the one-dimensional $\mathbb{F}_t$-space U_t/Z_t. It follows that the image contains all this additive group, and therefore equals Q_0/Z_q.

Thus $Q_0 = Z_q H$. The action of H on Z_q is exactly the scalar action of H_t, so W and every center intersection $L \cap Z_q$ are $\mathbb{F}_t$-subspaces. Conversely such a subspace V is normalized by H, making $Z_V H$ a subgroup. Writing every element as zh proves $L = (L \cap Z_q)H$. This gives (4), with the stated maximality criterion. □

This is an overgroup classification inside the specified mixed parabolic. It reduces maximality to one scalar-module dimension; no ambient Suzuki group enumeration or search over relative positions is required.

5 Uniform obstruction for Construction 10

Theorem D. For every parameter allowed by Construction 10, the intended bare subgroup-intersection lattice has a canonical embedding. No embedding having its prescribed mixed parabolic and pairwise mixed intersections is RWPRI.

Proof. Put $k = \alpha - 1 \geq 2$, select distinct odd relative primes a_i, and set

$$P = \prod_i a_i, \quad n = eP, \quad t = 2^e, \quad q = 2^n, \quad s_i = 2^{n/a_i}.$$

The source's bound $\alpha \leq \Omega(n)$ gives $\Omega(e) = \Omega(n) - k \geq 1$, hence $e > 1$. Choose canonical subfield groups $G_i = S_{n/a_i}$ and $Q = Z_q U_t H_t$. For every selection J of subfield types,

$$\bigcap_{j \in J} G_j = S_{n_J}, \qquad Q \cap \bigcap_{j \in J} G_j = Z_{2^{n_J}} U_t H_t. \tag{5}$$

These are exactly the intended field intersections and mixed-extension orders, including common Borel $U_t H_t$. The extension $Z_{2^{n_J}} U_t$ by its elementary abelian center kernel is nonsplit: every element outside that center has order four, so a nontrivial elementary abelian complement could not exist. Thus (5) also has the literal hat-dot type. This proves bare existence.

For any other proposed embedding, Lemma B puts its mixed parabolic into Q_0. Its pairwise intersections H_i have the prescribed type $(E_{s_i} \hat{\cdot} E_t) : C_{t-1}$. Their Sylow-2 groups cannot be elementary abelian: an extension of elementary abelian groups whose total group is elementary abelian splits as a vector-space extension. Lemma C therefore applies, and order division gives

$$|H_i \cap Z_q| = s_i, \qquad \dim_{\mathbb{F}_t}(\mathbb{F}_q/(H_i \cap Z_q)) = P - P/a_i.$$

Since $k \geq 2$ and every $a_i \geq 3$, this dimension exceeds one. Every H_i is therefore nonmaximal in Q_0. Explicitly, extend $H_i \cap Z_q$ by one $\mathbb{F}_t$-dimension; (4) gives a proper intermediate subgroup. Taking the source's RWPRI subset J consisting only of the mixed type leaves no maximal remaining intersection. This contradicts RWPRI. $\square$

The negative conclusion uses only the mixed parabolic and the specified pairwise intersection types. It does not require simultaneous alignment of a common simple subfield. It strengthens the earlier rank-three R12 argument in Earlier subfield and scalar proofs, Section C. It does not prove that arbitrary choices of the listed maximal parabolic types, with different intersections, cannot satisfy RWPRI. Nor does failure of RWPRI imply failure of the geometry or FT axioms.